# AICE 2009 Lab 1 Report

Rasul Abbaszada

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## Part A: FFT Basics

The Fast Fourier Transform of a generated sinusoidal signal was analyzed to observe its frequency components. The real part of the FFT corresponds to the cosine (even) components, while the imaginary part corresponds to the sine (odd) components, describing both amplitude and phase. The FFT of a real-valued signal displays conjugate symmetry around 0 Hz because the complex exponentials used in the transform must be in conjugate pairs to ensure their imaginary components cancel out during summation, resulting in purely real-valued signal. Main peaks were identified at $\pm 100$ Hz because this perfectly matches the input frequency of the generated sine wave. For a one-sided magnitude spectrum, the results were scaled by $2/N$. Dividing by $N/2$ normalizes the FFT sum into an average amplitude and accounts for the energy being split between positive and negative frequencies in a one-sided plot.

## Part B: Manual DFT vs. FFT

A manual Discrete Fourier Transform was implemented using a transformation matrix $W$ to compare with the FFT. Both methods produced identical magnitude spectra. The matrix $W$ acts as a set of complex basis sinusoids, where matrix multiplication compares the input signal against each frequency. In the heatmaps of $W$, the color patterns represent these oscillating values, with higher rows corresponding to higher frequencies and showing more frequent cycles. The computational complexity of the manual DFT is $O(N^2)$. The FFT reduces this to $O(N \log N)$. The frequency bin spacing is defined as $\Delta f = F_s/N$. Increasing the number of samples $N$ while keeping the sampling rate $F_s$ constant increases the total recording time, which decreases the bin spacing and improves the overall frequency resolution.

## Part C: FFT-Based Notch Filtering

A notch filter was used to remove a 600 Hz interference signal by creating a "hole" in the spectrum. A notch filter defines boundaries around a specific frequency to suppress bandpower. A strongly negative dB value indicates successful elimination of noise energy. The precision of targeting interference frequency is determined by the frequency bin spacing, $\Delta f = F_s/N$. If the interference frequency is not centered exactly on an FFT bin, spectral leakage occurs because energy spreads across neighboring bins due to the abrupt window edges. To mitigate this, windowing functions like Hann or Hamming are used to taper the signal edges smoothly. This reduces the leakage and confines the noise to a narrower range.

## Part D: Derivatives via FFT

Derivatives were calculated using both finite-difference and spectral FFT methods. The FFT spectral derivative provides a much closer match to the true analytical derivative than the finite-difference approach. This is because the finite-difference method relies on local approximations between neighboring points, which introduces truncation errors that are sensitive to grid spacing. In contrast, the FFT method is a global approach that analytically computes the derivative by representing the entire signal as complex sinusoids and multiplying the components by $i2\pi f$, entirely avoiding local approximation errors.